CRITICAL ANALYSIS OF POLICY RESEARCH: READY FOR COLLEGE? EXAMINING THE EFFECTIVENESS OF TARGETED INTERVENTIONS IN HIGH SCHOOL

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EXECUTIVE SUMMARY

This paper critically analyzes Xu et al. (2022), which evaluates Kentucky's Targeted Interventions (TI) program, a high school transition intervention triggered by 11th-grade ACT cutoff scores, to assess whether it improves students' early college outcomes and reduces reliance on remedial coursework. The central question is whether TI causally improves postsecondary outcomes, including remediation enrollment, enrollment, and success in credit-bearing courses, credit accumulation, and persistence. My critique focuses on how well the design supports causal claims, where bias may remain (especially noncompliance), and what additional analyses (e.g., IV/TOT estimation where feasible and deeper discussion of treatment-group composition) would strengthen policy conclusions about scaling TI-like interventions.

INTRODUCTION TO THE RESEARCH AREA

Postsecondary success remains a key policy priority, especially as college gaps persist across income and achievement lines. However, postsecondary graduation rates have been low and stagnating for the last decade. Only 30% of full-time community college students graduate within 3 years of initial enrollment.[1] Weak academic performance and preparation before college matriculation are considered key factors in blocking the path to student success in college. The state of Kentucky, in response to these challenges, adopted Common Core State Standards (CCSS)-aligned Kentucky Core Academic Standards (KCAS) in 2010 to get all students college-ready when graduating high school.[2] This included reforms in school curriculum, student assessments, school accountability, and most importantly, the introduction of a transition intervention program (TI).[3] This study evaluates the impact of TI, which aims to reduce the dependence on remedial coursework and developmental education in college, which delays degree attainment and disproportionately affects low-income and minority students.[4] This program uses 11th-grade American College Testing (ACT) exam scores to identify eligible

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[1] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 183

[2] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 184

[3] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 184

[4] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 185

students. Students scoring below 19 in Math, 18 in English, and 20 in Reading are considered eligible for the intervention.[5]

The core research question is: What are the effects of a high school-to-college transition intervention program in Kentucky on students' college outcomes?
The authors measure several outcomes, such as:

1. Remediation in math/English by the end of the first year
2. Enrollment in college-bearing courses
3. Passing a credit-bearing course and the total credits earned at the end of the first and second year.
4. Persistence to the second year of college

DATA AND METHODS USED:

This study utilizes longitudinal student-level administrative records provided by the Kentucky Centre for Statistics that track student progress from high school through college between 2009 and 2019.[6] These records include students' background characteristics, high school test scores, and interventions received. The sample of this study contains eight cohorts of 11th-grade students who took the ACTs between 2009 and 2016.[7] The unit of analysis is the individual student, and each observation corresponds to a student who took the mandatory 11th-grade ACT score between 2009 and 2016.[8] The authors implement a difference-in-regression-discontinuity (DiRD) design that estimates causal effects at a cutoff (E.g., an ACT score of 18) and compares outcomes before and after the policy change (the adoption of a TI program).
The regression equation is as follows:

$Yi= \alpha 0+\alpha 1Bi+k(Si)+\alpha 2Bi *k(Si)+\varepsilon i$

Where:

Yi= Independent variable (paper measures several outcomes, e.g. college enrollment)
$\alpha 0$ = Intercept
Bi= Binary indicator if a student scored below the ACT cutoff
$\alpha 1$= The jump in outcome at cutoff
Si = Student's ACT score centred around the cutoff
k(Si) = A control for the variable Si
Bi *k(Si) = Interaction term

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[5] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 185

[6] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 189

[7] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 189

[8] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg `190

εi = Error term[9]

The difference in discontinuity is as follows:
β^=α^1post−α^1pre , which is the change in the treatment group differenced by the change in the control group, and β^ is the effect attributable to the TI program.

SUMMARY AND DISCUSSION OF MAIN FINDINGS:

The study finds that the TI significantly reduces the likelihood of students enrolling in remedial math courses by 8 percentage points, which is a substantial improvement for students scoring near the ACT benchmark.[10] These effects are particularly stronger for Free and Reduced Price Lunch (FRL) students who saw a reduction in the need for remedial math by 10 percentage points.[11] This suggests that students eligible for the TI program are 8-10 percentage points less likely to be placed in remedial math compared to similar students before the intervention. This effect is consistent across both 2- and 4-year colleges.

TABLE 8
Estimated Effect of Targeted Interventions (TI) on College Outcomes

	Math				English			
	2-year		4-year		2-year		4-year	
Outcome	Coefficient/*SE*	Bandwidth/*N*	Coefficient/*SE*	Bandwidth/*N*	Coefficient/*SE*	Bandwidth/*N*	Coefficient/*SE*	Bandwidth/*N*
Remediation in subject by end of first year	−0.10*** (0.02)	4/55,396	−0.08*** (0.02)	4/63,301	−0.04 (0.02)	3/30,331	−0.03 (0.03)	2/20,172
Take credit-bearing course in subject by end of first year	−0.01 (0.04)	2/23,459	0.04 (0.02)	3/51,020	0.04 (0.02)	3/30,331	0.01 (0.02)	3/32,672
Take credit-bearing course in subject by end of second year	−0.00 (0.04)	2/23,459	0.03 (0.02)	3/51,020	0.04** (0.02)	3/30,331	−0.00 (0.01)	4/45,754
Pass credit-bearing course in subject by end of first year	0.00 (0.03)	3/42,087	0.04* (0.02)	4/63,301	0.01 (0.02)	4/40,736	−0.00 (0.03)	3/32,672
Pass credit-bearing course in subject by end of second year	0.02 (0.03)	3/42,087	0.04* (0.02)	4/63,301	0.03 (0.02)	4/40,736	−0.00 (0.03)	3/32,672

In addition to reducing math remedial enrollment, the study finds that TI increased enrollment in college-level credit-bearing courses in mathematics at 4-year institutions by 4 percentage points, and passing rates improved by a similar margin in these courses.[12] Among FRL students

[9] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 191

[10] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 193

[11] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 196

[12] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 201

attending 4-year universities, these gains were even more substantial, increasing the likelihood by 9 percentage points.[13] This suggests that students eligible for the TI program are 4-9 percentage points more likely to take credit-bearing mathematics courses than similar students before the intervention.

TABLE 11
Estimated Effect of Targeted Interventions (TI) on College Outcomes, by Student and School Characteristics

	Math				
	2-year		4-year		
Outcome	Coefficient/*SE*	Bandwidth/*N*	Coefficient/*SE*	Bandwidth/*N*	Coe
Students who missed ACT cutoff in addition to subject					
Remediation in subject by end of first year	−0.09*** (0.03)	4/39,538	−0.08*** (0.03)	4/32,158	−0.
Take credit-bearing course in subject by end of first year	−0.06 (0.05)	2/14,637	0.04 (0.03)	3/25,868	0.
Pass credit-bearing course in subject by end of first year	−0.03 (0.04)	3/28,695	−0.01 (0.03)	4/32,158	0.
Free/reduced-price lunch-eligible students					
Remediation in subject by end of first year	−0.10*** (0.04)	4/27,100	−0.11*** (0.04)	4/24,100	−0.
Take credit-bearing course in the subject by end of first year	−0.09 (0.07)	2/10,550	0.09** (0.05)	3/19,331	0.0
Pass credit-bearing course in subject by end of first year	−0.07 (0.05)	3/19,873	0.09** (0.04)	4/24,100	0.
Top quartile school in % students below cutoff					

However, TI showed no statistically significant effects on high school graduation rates or overall college enrollment. This indicates that while it may help students improve academically, it does not directly influence their decision to pursue post-secondary education. There was limited evidence of persistence or total credit accumulation by the end of the second year of college.[14] Even when estimated gains were positive, such as a 1.17 credit increase, they were considered statistically insignificant, likely due to small effects relative to variation.[15]
The authors estimate that TI effects are roughly equivalent to more than 0.20 SDs in the reduction in developmental math enrollment and 0.10 SDs in the increase of taking and passing college math, which are considered sizable effects relative to the cost of the program. For instance, according to an earlier study, the per-student cost of TI was estimated to be $600 as opposed to each developmental course in college costing $830 in community colleges.[16]

CRITICAL ASSESSMENT OF THE ANALYSIS:
1) Validity of the Research Design:

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[13] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 201
[14] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 196

[15] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 196
[16] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 204

The authors rightfully dismiss the naive OLS regression that estimates the association between participation in TI and student outcomes. The following equation depicts an OLS regression model:

$Y_i = \beta_0 + \beta_1 \cdot TI_i + \varepsilon_i$

Where:

Y_i= Individual student outcome, such as enrollment in college

TI_i = Indicator for TI (=1 if the student participated in TI, 0 otherwise)

β_0 = The average outcome for students who did not receive TI.

β_1 = The difference in average outcomes between students who received TI and those who did not.

However, this approach is inadequate due to three main sources of bias:

- Selection Bias: The intervention is not randomly assigned, and students who are eligible for the program differ systematically from those who are not, particularly because participation depends on ACT benchmark scores. An OLS regression would overstate the negative outcomes for TI participants, conflating the program's effects with pre-existing disadvantages.
- Possible confounding results: The same ACT cutoffs utilised to determine eligibility are utilised to determine eligibility for developmental education in college.[17] Thus, students below the cutoff are subject to both high-school interventions and remediation in college. A naive regression analysis cannot disentangle the effects of the intervention from pre-existing policies and may incorrectly attribute effects on college outcomes to TI when they could be due to college remediation courses.
- Differences in Baseline Ability: An OLS estimate inaccurately compares all students who received TI with those who did not, regardless of how close or how far their scores are from the benchmarks. This leads to comparisons between students with very different baseline academic strengths and varying degrees of college readiness. This would further confound the estimated effect of the intervention on underlying academic ability.

Why a Standard RD is also Insufficient: A natural alternative to an OLS estimation is regression discontinuity (RD) analysis, which compares students just above and below the cutoff scores. These students are more academically similar, and the cutoff provides a random assignment mechanism. However, an RD also fails in this setup because the same cutoff score (E.g. ACT score in math < 19) is a trigger for two different policies:

1. For all cohorts, both pre-and post, students below the cutoff are more likely to be placed in remedial classes in college.
2. Only students in the post-TI cohorts, students below the cutoff, are offered an alternative to the TI program.[18]

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[17] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 190

[18] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 190

Due to this, the RD result cannot disentangle the effects of the two policies: TI and the continuing effect of the college placement policy. Without adjusting for the pre-existing discontinuity caused by the remediation policy, the RD also risks misattributing outcomes to the high school intervention. For post-TI cohorts, the discontinuity in outcomes at the cut-off in an RD set-up would represent the combined effect of TI and college placement policy.[19] This scenario calls for the use of the DiRD approach.

To overcome this issue, the DiRD design compares the size of the discontinuity in outcomes at the ACT cutoff before and after the TI was implemented.[20] In the pre-TI cohort, the effect is captured by remedial placement policies; in the post-TI cohort, the effect of both policies in action (TI and placement) is captured. Subtracting the pre-TI discontinuity from the post-TI discontinuity isolates the effect of the TI alone.
Effect of TI = Jumppost - Jumppre. The pre-TI discontinuity was estimated to be 16 percentage points; this means that in the pre-TI period, students scoring just below the ACT math cutoff were 16 percentage points more likely to enroll in college remedial math compared to students scoring just above the cutoff. The post-TI discontinuity was estimated at 6 percentage points, meaning that post-TI cohort students scoring just below the ACT math cutoff were 6 percentage points more likely to enrol in college remedial math.

Putting the DiRD logic in action, this means that:
The effect of TI = 6 - 16 = -10 percentage points. This 10 percentage-point reduction can be attributed to the effect of TI alone, because the effect of the college placement policy has been differenced out. This approach enables the authors to make a stronger causal claim, as they are not just comparing TI students to non-TI students but also measuring how the treatment effect at the cutoff changed after TI was introduced, while holding other factors constant.

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[19] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209As a result. Pg 190
[20] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209: 100% of students below the cutoff score receive the intervention, and 0% of those above the cutoff . Pg 191

TABLE 9

Robustness of Results to Alternate Bandwidths

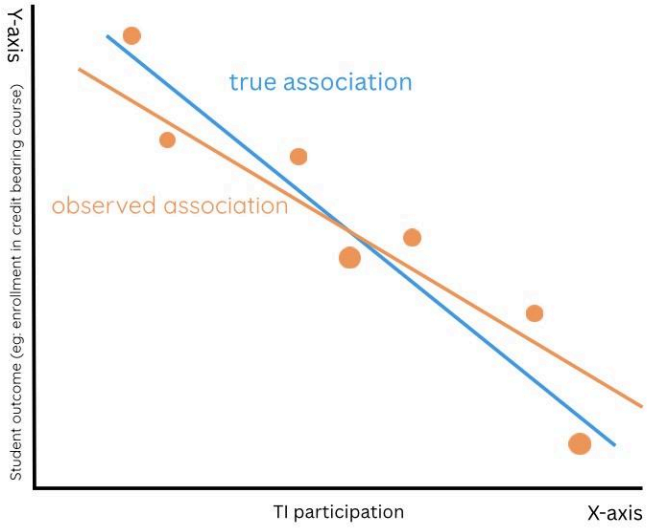
	Math					
	2-year			4-year		
Outcome	BW	BW + 1	BW + 2	BW	BW + 1	B
Take remedial course in subject by end of first year						
Discontinuity pre	0.16***	0.19***	0.21***	0.10***	0.11***	0
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	
Discontinuity post	0.06***	0.08***	0.10***	0.02***	0.03***	0
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	
Difference in RD	−0.10***	−0.11***	−0.10***	−0.08***	−0.08***	−(
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	

2) Attenuation Bias in DiRD design

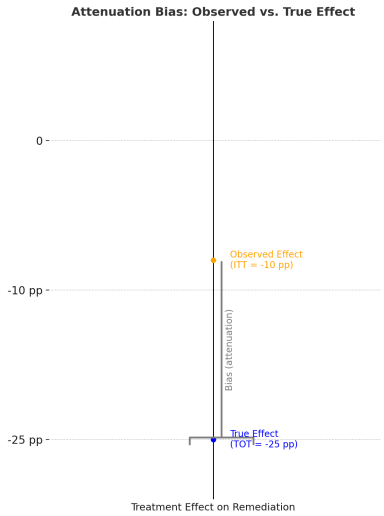
The treatment design is not sharp in this setup. As shown in the table, only 60% of students in the treatment group received the intervention, and up to 30% of those above the cutoff also received it. A foundational assumption of DiRD and RD is that treatment assignment is sharp, and the expectation is that 100% of the students below the cutoff score received the intervention, and 0% above should not. This non-compliance yields a fuzzy RD that biases estimates towards zero, leading to an attenuation bias. The authors acknowledge this effect, stating that the discontinuity begins at around 40 percentage points in both subjects, and surveys conducted with TI administrators indicate teacher input in conjunction with ACT scores as factors to refer students to TI.[21] Thus, the authors estimate the Intent-to-Treat (ITT) effect, not the Treatment-on-the-Treated (TOT) effect. This muddies the treatment group, and the key idea here is that the true effect of TI on students who received it (TOT) is likely to be larger than the reported effect.

Using the paper's numbers, it is possible to estimate the magnitude of the attenuation bias utilising the ITT estimates and the observed compliance rate of 40 percentage points.
The magnitude of attenuation can be estimated as:
TOT = ITT/Compliance Rate
Bias = Estimated TOT - ITT
Based on the paper's findings, the ITT's effect on math remediation enrollment is estimated at a 10 percentage-point reduction.
Compliance rate = 0.40
ITT = -10 percentage point
TOT = -10/0.40 = -25
Attenuation Bias = -25 - (-10) = -15

[21] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 186



Based on this calculation, we can say that the ITT estimate understates the treatment effect by 15 percentage points, resulting in a downward bias. It is noteworthy to mention, that the authors performed a back-of-the-envelope calculation which also suggests that the true reduction in remedial math enrolment is higher and estimated at 20-25 percentage points (similar to the TOT calculation performed in this paper), and the increase in passing credit-bearing courses in math could be 10 percentage points as compared to the ITT effect that estimated it to be around 4 percentage points.[22]



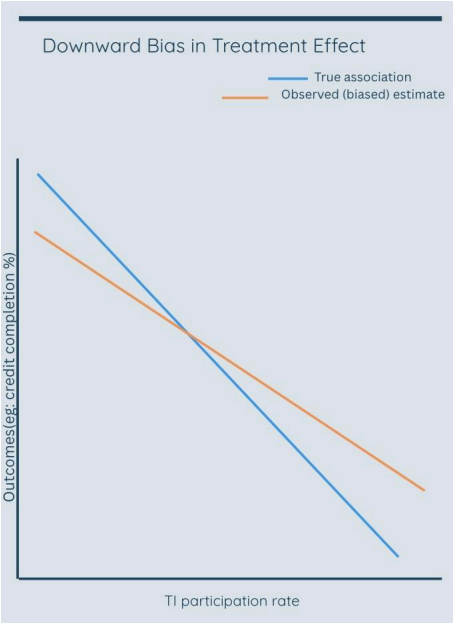
3) Change in Composition of Treated Students

An important limitation of the DiRD approach in this paper is the potential for compositional bias. The DiRD design assumes that the effect of scoring below the cutoff is comparable across

[22] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 198

all cohorts, which means they are assuming that the composition of students placed into college remediation is similar before and after the TI. Before the introduction of TI, students scoring below the cutoff may have included moderately to severely underprepared students who needed remediation. The moderately underprepared students may have been more likely to succeed in college-level coursework if given the proper support. However, with the introduction of TI, marginally underprepared students who might have previously gone into remediation are receiving support in high school, potentially avoiding remediation altogether. As a result, those who are still placed in remediation post-TI are the hardest to help students with severe academic struggles. This creates a compositional shift in who is treated, even though all students are below the 'cutoff'.

As the post-TI remediation group is more academically disadvantaged, their college outcomes, such as enrolling and passing in credit-bearing courses, appear to be worse. This means that even if TI is effective, its effects appear smaller in the DiRD estimate, leading to a downward bias. The key idea here is that TI's effects are understated because the treatment group appears worse, and not because the intervention is ineffective.



INSIGHTS GAINED FROM THE STUDY:

This paper strengthened my understanding of the DiRD design and how this approach can be utilized to isolate policy effects over time when RD approaches are insufficient. The authors address a policy-relevant question about the effectiveness of TI initiatives in improving school and college outcomes using a robust research design. The authors address the fact that there is less research in this area, and their research adds to the body of evidence regarding the success of TI programs as a cost-effective alternative to supplemental courses in college.[23] They are also

[23] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 204

transparent about the limitations of their paper and employ robustness checks, sensitivity analysis, and balance tests to demonstrate the validity of their findings. By applying an equity lens and disaggregating results by income level and school quality, they highlighted how interventions may affect different student populations.

However, the paper also exposed limitations, particularly the reliance on ITT rather than TOT estimates, due to imperfect compliance and the lack of participation data in pre-TI years. This means the program's true effects are likely understated, as evidenced by critiques that indicate a downward bias in estimates. The authors could have used an Instrumental Variables approach to estimate TOT in the post-TI cohorts, where participation data is available. Utilising the ACT score cutoff as a valid instrument would offer a clearer picture of the intervention's efficacy among actual participants. Secondly, the study could have acknowledged the compositional changes in the treated group, and a rigorous discussion of this aspect in their papers can strengthen their causal claims.

Future research should prioritize understanding compliance rates and the timing of intervention delivery, as highlighted by the authors.[24] The study suggests that critical college-going decisions are made as early as middle school, and that students who fall below readiness benchmarks in 9th grade often struggle to catch up by graduation.[25] These findings call for earlier, more targeted interventions. Overall, this paper underscores the importance of rigorous causal attention, attention to program delivery, and emphasizes the equity implications of initiatives that help students succeed beyond high school.

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[24] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 204

[25] Xu, Z., Backes, B., Oliveira, A., & Goldhaber, D. (2022). Ready for college? examining the effectiveness of targeted interventions in high school. *Educational Evaluation and Policy Analysis*, *44*(2), 183–209. Pg 204